

Paper Practice 2.1**Writing and Interpreting Equations**

Show all of your work in the space provided. Define every variable you use. Circle your final answers.

- 1.** Translate each sentence into an equation.
Use n for the number. Do NOT solve.
(Translate)

a. The sum of a number and 12 is 30.

b. Seven less than twice a number is 19.

c. The product of 6 and a number, decreased by 4, equals 20.

d. Three-fourths of a number is 27.

- 2.** Translate each equation into a sentence.
(Translate)

a. $5n + 3 = 28$

b. $n \div 4 - 2 = 9$

c. $2(n + 6) = 40$

- 3.** A gym charges a \$25 sign-up fee plus \$18 per month. Marisol has paid \$169 in total. Define a variable and write an equation. Do NOT solve. (Write)

- 4.** A rectangle's length is 4 cm more than its width, and its perimeter is 56 cm. Define a variable and write an equation for the perimeter. (Write)

- 5.** Devon bought 3 shirts at the same price and a \$12 hat, spending \$57. Define a variable and write an equation. (Write)

- 6.** Is the given value a solution? Substitute and show the check. (Check)

a. $4x - 9 = 15$; $x = 6$

b. $3(x + 2) = 27$; $x = 8$

c. $x \div 2 + 5 = 8$; $x = 6$

- 7.** The equation $45 + 0.20m = 61$ models a one-day car rental bill. Tell what each part represents: 45, 0.20, m , and 61. (Interpret)

- 8.** The equation $8h + 6.50 = 54.50$ models Kai's earnings for one day. Write a real-world situation that fits it. (Interpret)

- 9.** Error Analysis — Gus translated “six less than a number is 14” as $6 - n = 14$. Identify his mistake, then write the correct equation. (Error Analysis)

- 10.** The perimeter of a square is $P = 4s$. Write the equation for a square with perimeter 62 cm, then find s using mental math. (Apply)

- 11.** Concert tickets cost \$32 each, plus a \$9 service fee per order. (Write)

a. Write an equation for the total cost C of an order of t tickets.

b. Write the equation you would solve to find how many tickets \$137 buys.

c. Is $t = 4$ a solution of your equation in part b? Check it.

- 12.** Always, sometimes, or never true? Explain each. (Reason)

a. “A number increased by 5 equals 5 increased by the number.” $n + 5 = 5 + n$

b. “A number decreased by 5 equals 5 decreased by the number.” $n - 5 = 5 - n$